

ADVANCED CRM + HRMS + ERP

Functional & Technical Implementation Specification

CRM • Sales • HRMS • Payroll • Accounting • Ledger • Inventory • Projects • Helpdesk • Automation • AI

Document Type	Software Requirements & Implementation Blueprint
Version	1.0
Target	Web-based multi-company / multi-branch business platform
Architecture	Modular monolith initially; service-ready boundaries
Primary Users	Management, Sales, HR, Finance, Operations, Employees, Customers, Vendors
Status	Implementation-ready baseline specification

1. Purpose and Product Vision

This document defines the functional scope, system architecture, data model, workflows, security controls, APIs, implementation phases, testing strategy and deployment requirements for an advanced business management application combining CRM, HRMS, ERP operations and double-entry accounting.

1.1 Product Objectives

- Maintain a single source of truth for leads, customers, vendors, employees, products, projects and financial transactions.
- Automate lead-to-cash, procure-to-pay, hire-to-retire, attendance-to-payroll and ticket-to-resolution workflows.
- Provide auditable double-entry accounting with customer/vendor ledgers, receivables, payables and financial statements.
- Support role-based, branch-based and company-based data access.
- Provide management dashboards, alerts, scheduled reports, workflow approvals and configurable automation.
- Expose integration-ready REST APIs and webhooks for external systems.

1.2 Recommended Delivery Strategy

Build the platform in controlled releases. Establish identity, authorization, master data and audit logging first; then CRM/Sales, HRMS, Finance, Operations and finally advanced automation/AI. Financial posting rules and payroll calculations should be independently testable domain services.

2. User Roles and Access Model

Role	Typical Access
Super Admin	Tenant/company provisioning, global settings, modules, security and audit.
Company Admin	Company configuration, branches, users, workflows and reports.
Sales Manager	Leads, pipeline, targets, quotations, approvals and sales reports.
Sales Executive	Assigned leads/customers, activities, follow-ups, quotations.
HR Admin	Employees, attendance, leave, payroll, recruitment and HR reports.
Finance/Accountant	Chart of accounts, vouchers, ledgers, banking, tax and financial reports.
Operations Manager	Projects, inventory, purchases, vendors, tasks and helpdesk.
Employee	ESS, attendance, leave, payslips, tasks, expenses and timesheets.
Customer	Portal access to quotations, invoices, payments, projects and tickets.
Vendor	Portal access to POs, bills, payment status and documents.

2.1 Authorization Requirements

- RBAC permissions at module/action level: view, create, edit, delete, approve, export and administer.

- Data-scope rules: own records, team, department, branch, company or all permitted records.
- Field-level restrictions for sensitive salary, bank, tax and commercial information.
- Maker-checker controls for financial approvals and configurable approval limits.
- Every privileged action must generate an immutable audit event.

3. High-Level System Architecture

Recommended initial architecture: modular monolith with clearly separated domain modules and a shared platform layer. This reduces operational complexity while retaining boundaries for later extraction into services.

Layer	Responsibilities
Presentation	Responsive web application, dashboards, forms, portals, accessibility.
API/Application	Authentication, validation, use cases, orchestration, permissions.
Domain	CRM, Sales, HRMS, Payroll, Accounting, Inventory, Projects, Helpdesk rules.
Persistence	Relational database, object/file storage, cache and search indexes.
Async/Jobs	Notifications, scheduled reports, imports, reminders, payroll jobs, webhooks.
Integration	Email/SMS/WhatsApp, payment gateway, biometric, GST/e-invoice readiness.
Observability	Central logs, audit logs, metrics, traces, health checks and alerting.

3.1 Suggested Technology Baseline

- Frontend: React/Next.js or equivalent component-based web framework.
- Backend: Node.js/NestJS, Java/Spring Boot, .NET, or Laravel depending on team capability.
- Database: PostgreSQL recommended for transactional integrity and reporting.
- Cache/queue: Redis plus a durable background-job mechanism.
- Files: S3-compatible object storage with signed access URLs.
- Authentication: secure session/JWT implementation with refresh rotation and MFA support.
- Reporting: server-side report service for PDF/Excel exports and scheduled delivery.

4. Core Platform Modules

4.1 Dashboard & Analytics

- Role-specific dashboards
- Configurable KPI cards
- Date/branch/company filters
- Drill-down charts
- Pending approvals
- Alerts and reminders
- Saved dashboard views

4.2 User, Role & Permission Management

- Users and status lifecycle
- Roles and permission matrix
- Data scopes
- MFA support
- Session/device management
- Password policy
- Login history

4.3 Company & Branch Management

- Multiple companies
- Multiple branches
- Financial years
- Company branding
- Branch addresses/GST details
- Inter-branch controls
- Consolidated reporting

4.4 Master Data Management

- Departments/designations
- Locations
- Currencies
- Units
- Tax codes
- Payment terms
- Numbering sequences
- Categories/tags
- Custom fields

4.5 Document Management

- Central attachments
- Entity-linked documents
- Document categories
- Expiry reminders
- Version metadata
- Permission-controlled download
- Virus scanning hook

4.6 Notification Engine

- In-app notifications
- Email/SMS/WhatsApp adapters
- Template variables
- Event-driven notifications
- Scheduled reminders
- Delivery status
- User preferences

4.7 Workflow & Approval Engine

- Configurable states
- Multi-level approvals
- Conditional routing
- Amount thresholds
- Escalations
- Delegation
- Approval history
- SLA timers

4.8 Audit & Activity Logs

- Who/what/when metadata
- Before/after values for critical changes
- Login/security events
- Export events
- Approval events
- Financial posting events
- Searchable audit console

5. CRM and Sales Modules

5.1 Lead Management

- Lead creation/import/API capture
- Source/campaign tracking
- Owner/team assignment
- Custom lead stages
- Lead scoring
- Tags/custom fields
- Duplicate detection
- Notes/files
- Lead aging
- Lost reasons
- Conversion to customer/contact/opportunity

5.2 Customer & Contact Management

- Organization and individual customers
- Multiple contacts
- Addresses
- Tax/KYC fields
- Credit limit
- Customer tags/segments
- Interaction timeline
- Files
- Outstanding summary
- Customer statement

5.3 Opportunity / Deal Pipeline

- Multiple pipelines
- Stage probability
- Deal value
- Expected close date
- Products/services
- Competitors
- Activities
- Win/loss reason
- Forecast category
- Pipeline analytics

5.4 Activities & Follow-ups

- Calls
- Meetings
- Tasks
- Reminders
- Recurring activities
- Calendar
- Outcome codes
- Overdue follow-ups
- Activity timeline

5.5 Product & Service Catalogue

- SKU/service code
- Category
- UOM
- Price lists
- Taxes
- HSN/SAC
- Active/inactive
- Customer-specific pricing
- Discount rules

5.6 Quotations

- Quote numbering
- Line items
- Taxes/discounts
- Terms
- Validity
- Approval
- PDF generation
- Revision history
- Customer acceptance
- Convert to sales order

5.7 Sales Orders

- Order confirmation
- Delivery/billing details
- Order status
- Partial fulfillment
- Invoice conversion
- Cancellation controls

5.8 Invoicing & Collections

- Tax invoices
- Recurring invoices
- Credit/debit notes
- Partial payment
- Advance receipt
- Payment allocation
- Due dates
- Aging
- Collection reminders
- Customer statement

6. HRMS Modules

6.1 Employee Master

- Employee profile
- Employee ID
- Department/designation
- Manager hierarchy
- Employment type
- Joining/exit
- Bank and statutory fields
- Emergency contact
- Documents
- Lifecycle status

6.2 Attendance

- Check-in/out
- Shift mapping
- Web/mobile attendance
- Biometric integration interface
- GPS/geofence optional
- Late/early rules
- Overtime
- Regularization
- Attendance lock

6.3 Shift & Roster

- Shift templates
- Weekly offs
- Roster assignment
- Grace periods
- Night shift
- Overtime rules
- Shift swaps

6.4 Leave Management

- Leave types
- Accruals
- Opening balance
- Applications
- Half day
- Comp-off
- Multi-level approval
- Holiday calendar
- Encashment
- Team calendar

6.5 Payroll

- Salary structures
- Earnings/deductions
- Attendance inputs
- LOP
- Overtime
- Bonus/incentive
- Loans/advances
- Reimbursements
- PF/ESI/PT/TDS configuration
- Payroll run
- Payslip
- Bank advice
- Payroll lock
- Full & final settlement

6.6 Recruitment / ATS

- Job requisitions
- Openings
- Candidates
- Resume storage
- Pipeline stages
- Interviews
- Scorecards
- Offer letters

- Joining
- Candidate-to-employee conversion

6.7 Performance

- KPI/KRA
- Goals
- Review cycles
- Self review
- Manager review
- Ratings
- 360 feedback optional
- PIP
- Promotion recommendations

6.8 Employee Self-Service

- Profile
- Attendance
- Leave
- Payslips
- Tax/payroll documents
- Expenses
- Timesheets
- Tasks
- HR requests
- Announcements

7. Accounting, Ledger and Finance

The accounting subsystem must use double-entry bookkeeping. Business documents such as invoices, bills, receipts and payments should post balanced journal entries through a controlled posting service. Posted entries should not be silently edited; corrections should use reversal/adjustment transactions.

7.1 Chart of Accounts

- Account groups
- Assets/liabilities/equity/income/expense
- Control accounts
- Company-specific COA
- Opening balances
- Active/inactive controls

7.2 General Ledger

- Journal entries
- Posting date
- Document date
- Reference
- Narration

- Debit/credit lines
- Cost center
- Branch
- Project
- Source document
- Posting status

7.3 Vouchers

- Receipt voucher
- Payment voucher
- Journal voucher
- Contra voucher
- Debit note
- Credit note
- Numbering
- Approval
- Attachment
- Posting/reversal

7.4 Accounts Receivable

- Customer control account
- Invoice-wise outstanding
- Advances
- Allocation
- Aging buckets
- Credit limit
- Statements
- Collection follow-ups

7.5 Accounts Payable

- Vendor control account
- Bill-wise outstanding
- Advances
- Payment schedule
- Aging
- Vendor statements
- Payment approvals

7.6 Banking & Cash

- Multiple bank/cash accounts
- Transfers
- Deposits/withdrawals
- Bank reconciliation
- Cheque/reference tracking
- Cash book
- Bank book

7.7 Expense Management

- Expense categories
- Claims
- Bills/receipts
- Approval
- Reimbursement
- Petty cash
- Department/project/cost-center allocation

7.8 Tax/GST

- GSTIN per registration
- CGST/SGST/IGST logic
- HSN/SAC
- Place of supply
- Tax-inclusive/exclusive pricing
- TDS configuration
- Tax reports
- E-invoice/e-way bill integration readiness

7.9 Financial Statements

- Trial balance
- Profit & Loss
- Balance Sheet
- Cash flow
- General ledger
- Day book
- Receivable aging
- Payable aging
- Cost-center reports
- Branch/company consolidation

7.10 Example Accounting Postings

Business Event	Debit	Credit
Sales invoice	Accounts Receivable	Sales Revenue + Output Tax
Customer receipt	Bank/Cash	Accounts Receivable
Purchase bill	Expense/Inventory + Input Tax	Accounts Payable
Vendor payment	Accounts Payable	Bank/Cash
Payroll accrual	Salary/Payroll Expense	Payroll Payable
Salary payment	Payroll Payable	Bank
Customer credit note	Sales Return/Adjustment	Accounts Receivable

8. Purchase, Vendor and Inventory

8.1 Vendor Management

- Vendor profile
- Contacts

- GST/tax data
- Bank details
- Documents
- Payment terms
- Outstanding
- Performance

8.2 Purchase Management

- Purchase requisition
- RFQ
- Vendor quotation
- Comparison
- Purchase order
- Approval
- Goods receipt
- Purchase bill
- Purchase return
- Payment linkage

8.3 Inventory

- Items/SKUs
- Warehouses
- Opening stock
- GRN
- Issue/outward
- Transfers
- Adjustments
- Reorder level
- Low-stock alerts
- Batch/serial optional
- Stock valuation
- Stock ledger
- Physical count

9. Project, Task, Timesheet and Helpdesk

9.1 Project Management

- Projects
- Client linkage
- Members
- Milestones
- Budget
- Status
- Billing model
- Files

- Project expenses
- Project profitability

9.2 Task Management

- Tasks/subtasks
- Assignee
- Priority
- Status
- Dependencies
- Due dates
- Comments
- Attachments
- Recurring tasks
- Kanban/list/calendar

9.3 Timesheets

- Time entries
- Project/task linkage
- Billable/non-billable
- Approval
- Locking
- Utilization reports

9.4 Helpdesk

- Tickets
- Categories
- Priority
- SLA
- Assignment
- Escalation
- Internal notes
- Customer replies
- Attachments
- Resolution
- CSAT
- Ticket analytics

10. Portals, Communication, Automation and AI

10.1 Customer Portal

- Secure login
- Profile
- Quotations
- Invoices
- Payments

- Statements
- Projects
- Tickets
- Documents

10.2 Vendor Portal

- Profile
- Purchase orders
- Bills
- Payment status
- Documents
- Queries

10.3 Communication

- Email integration
- SMS adapter
- WhatsApp Business integration adapter
- Templates
- Conversation/activity logging
- Opt-in/consent fields

10.4 Automation

- Trigger-event-condition-action rules
- Auto assignment
- Reminder rules
- Scheduled reports
- Document expiry
- Low-stock alerts
- Lead inactivity
- Payment reminders
- Webhooks

10.5 AI Features - Optional Advanced Release

- Lead scoring assistance
- Conversation summaries
- Follow-up suggestions
- Draft email/message assistance
- Natural-language report query
- Forecast summaries
- Anomaly flags
- Knowledge assistant with permission-aware retrieval

11. Core Data Model

Use UUIDs or equivalent non-guessable identifiers for externally exposed records. Every business table should normally include `company_id`, `branch_id` where applicable, `created_at`, `updated_at`, `created_by`, `updated_by` and a status/soft-delete strategy.

Domain	Core Entities
Identity	users, roles, permissions, role_permissions, user_roles, sessions
Organization	companies, branches, departments, designations, cost_centers
CRM	leads, lead_sources, contacts, customers, opportunities, activities, notes
Sales	products, services, price_lists, quotations, quotation_items, sales_orders, invoices, invoice_items, receipts
HRMS	employees, shifts, attendance, leave_types, leave_requests, salary_structures, payroll_runs, payslips
Recruitment	jobs, candidates, applications, interviews, offers
Finance	accounts, account_groups, journal_entries, journal_lines, vouchers, bank_accounts, reconciliations
Purchasing	vendors, purchase_requisitions, rfqs, purchase_orders, grns, purchase_bills
Inventory	items, warehouses, stock_transactions, stock_balances, stock_counts
Projects	projects, milestones, tasks, timesheets, project_members
Helpdesk	tickets, ticket_messages, sla_policies
Platform	documents, notifications, workflows, approvals, audit_logs, webhooks, custom_fields

12. Critical Workflows

12.1 Lead-to-Cash

Lead → Qualification → Opportunity → Quotation → Approval → Customer Acceptance → Sales Order → Invoice → Payment → Receipt Allocation → Ledger Posting → Customer Statement.

12.2 Procure-to-Pay

Purchase Requisition → Approval → RFQ → Vendor Selection → Purchase Order → Goods Receipt → Purchase Bill → Approval → Vendor Payment → Ledger Posting.

12.3 Hire-to-Retire

Job Opening → Candidate → Interview → Offer → Employee Creation → Attendance/Leave → Payroll → Performance → Exit → Full & Final Settlement.

12.4 Attendance-to-Payroll

Shift/Roster → Attendance Capture → Regularization → Attendance Lock → Payroll Inputs → Payroll Calculation → Approval → Payslips → Salary Payment → Accounting Posting.

12.5 Ticket-to-Resolution

Ticket Creation → Categorization → Assignment → SLA Tracking → Escalation → Resolution → Customer Confirmation/Closure → CSAT.

13. API Design

Use versioned REST APIs initially. APIs must enforce authorization server-side regardless of UI visibility.

Area	Example Endpoints
Auth	POST /api/v1/auth/login; POST /refresh; POST /logout
Leads	GET/POST /api/v1/leads; GET/PATCH /leads/{id}; POST /leads/{id}/convert
Customers	GET/POST /customers; GET /customers/{id}/statement
Quotes	POST /quotations; POST /quotations/{id}/approve; POST /quotations/{id}/convert
Invoices	POST /invoices; POST /invoices/{id}/post; POST /invoices/{id}/credit-note
Employees	GET/POST /employees; PATCH /employees/{id}
Attendance	POST /attendance/check-in; POST /check-out; POST /regularizations
Leave	POST /leave-requests; POST /leave-requests/{id}/approve
Payroll	POST /payroll-runs; POST /payroll-runs/{id}/calculate; POST /approve
Ledger	GET /accounts/{id}/ledger; POST /journal-entries; POST /journal-entries/{id}/post
Reports	GET /reports/trial-balance; /profit-loss; /balance-sheet; /aging
Files	POST /documents; GET /documents/{id}/download
Webhooks	POST /webhooks; GET /webhooks/deliveries

13.1 API Standards

- JSON request/response contracts
- Idempotency keys for payment/posting endpoints
- Pagination/filter/sort standards
- Validation error schema
- Rate limiting
- Request correlation ID
- OpenAPI specification
- Webhook signatures and retry policy

14. UI / UX Requirements

- Responsive desktop-first administration interface with usable tablet/mobile ESS and portals.

- Global search for permitted customers, leads, employees, invoices and projects.
- Consistent list pages: search, filters, saved views, column chooser, pagination and export.
- Entity detail pages should use summary header + tabs + activity timeline.
- Forms require inline validation, unsaved-change protection and permission-aware controls.
- Financial screens must clearly distinguish Draft, Approved, Posted, Reversed and Cancelled states.
- Accessibility: keyboard navigation, labels, contrast, focus states and semantic form controls.

15. Security and Compliance Requirements

- TLS for all network traffic; secure password hashing; secrets stored outside source code.
- MFA for privileged roles and optional enforcement by company.
- OWASP-aligned input validation, output encoding, CSRF/session protections and secure headers.
- Parameterized database access and least-privilege database credentials.
- Encryption for sensitive stored fields where appropriate; encrypted backups.
- Object storage access via time-limited signed URLs.
- Audit trail for authentication, permissions, exports, approvals, payroll and accounting.
- Tenant/company isolation tests must be mandatory.
- Configurable data retention and account deactivation procedures.
- Regular dependency scanning, vulnerability testing and backup restoration drills.

16. Non-Functional Requirements

Area	Baseline Requirement
Availability	Target defined per deployment; production health checks and graceful recovery.
Performance	Typical list/detail API responses targeted under 2 seconds at normal load.
Scalability	Stateless application nodes; background workers independently scalable.
Reliability	Transactional posting for accounting/payroll critical operations.
Auditability	Critical changes attributable to user, timestamp, source and affected record.
Backup	Automated database/file backups with tested restore procedure.
Observability	Structured logs, error tracking, metrics and alerting.
Localization	Configurable timezone, date format, currency and tax settings.
Export	PDF/Excel/CSV where business appropriate.
Accessibility	WCAG-oriented controls for core workflows.

17. Reporting Catalogue

Module	Key Reports
CRM	Lead source, conversion, aging, pipeline, win/loss, salesperson performance
Sales	Quotation, order, invoice, collection, revenue,

	product/service sales
HRMS	Headcount, attendance, leave, overtime, attrition, recruitment
Payroll	Payroll register, payslip, deductions, bank advice, department cost
Finance	Day book, ledger, trial balance, P&L, balance sheet, cash flow, aging
Purchase	PO, vendor purchase, outstanding, purchase analysis
Inventory	Stock ledger, valuation, movement, reorder, aging
Projects	Progress, utilization, timesheet, cost, billing, profitability
Helpdesk	Ticket volume, SLA, resolution time, agent performance, CSAT

18. Implementation Roadmap

Phase	Scope	Indicative Outcome
Phase 0	Architecture, UX system, CI/CD, environments, coding standards	Development foundation
Phase 1	Auth, users, RBAC, companies, branches, masters, audit, documents	Core platform
Phase 2	Leads, customers, opportunities, activities, products, quotations	CRM MVP
Phase 3	Sales orders, invoices, receipts, AR and basic ledger posting	Lead-to-cash
Phase 4	Employees, attendance, shifts, leave, ESS	HRMS core
Phase 5	Payroll, statutory configuration, payroll accounting	Payroll
Phase 6	COA, vouchers, AP, banking, reconciliation, financial statements	Finance/ledger
Phase 7	Vendors, purchasing, inventory	ERP operations
Phase 8	Projects, tasks, timesheets, helpdesk, portals	Operations & portals
Phase 9	Workflow builder, automation, integrations, scheduled reports	Automation
Phase 10	AI assistance, advanced analytics, optimization	Advanced release

19. Testing Strategy

- Unit tests for domain rules: tax, discount, attendance, leave, payroll and journal balancing.
- Integration tests for database transactions, queues, files and external adapters.
- API contract tests and authorization tests for every protected endpoint.
- Accounting invariant tests: total debits must equal total credits for every posted journal.
- Payroll golden-data tests using approved expected calculations.
- End-to-end tests for lead-to-cash, procure-to-pay and attendance-to-payroll.
- Tenant/branch isolation and privilege-escalation tests.

- Performance/load tests for dashboards, large ledgers, payroll and report exports.
- Backup/restore and disaster-recovery tests before production.
- UAT scripts with sign-off by Sales, HR, Finance and Operations stakeholders.

20. Deployment and DevOps

- Separate development, staging/UAT and production environments.
- Containerized deployment recommended with environment-specific configuration.
- CI pipeline: lint → unit tests → security checks → build → integration tests → deploy.
- Database migrations must be version-controlled, reversible where feasible and backed up before risky changes.
- Use rolling/blue-green deployment strategy when scale requires minimal downtime.
- Centralized logs and error monitoring; production alerts for failed jobs, high error rate and resource saturation.
- Scheduled database and object-storage backups with retention policy.

21. Definition of Done

- Functional requirements implemented and acceptance criteria passed.
- Permissions verified for allowed and denied roles.
- Audit events present for critical actions.
- Unit/integration tests passing and required coverage threshold met.
- No unresolved critical/high security defects.
- Responsive UI and validation verified.
- Reports reconcile to source transactions.
- Accounting postings balance and cannot be silently mutated after posting.
- Technical/API documentation updated.
- UAT approval completed before production release.

22. Recommended MVP

For the first commercially usable release, avoid implementing all advanced features simultaneously. The recommended MVP is:

1. Core platform: users, roles, permissions, company, branch, masters, documents and audit.
2. CRM: leads, customers, opportunities, follow-ups and activities.
3. Sales: products/services, quotations, invoices, payments and customer outstanding.
4. HRMS: employees, attendance, leave and basic ESS.
5. Finance: chart of accounts, journal/vouchers, customer/vendor ledgers, cash/bank, trial balance and P&L.
6. Dashboard, notifications, standard reports and Excel/PDF exports.

Payroll, inventory, purchasing, projects, helpdesk, portals, workflow builder, advanced integrations and AI can then be added as controlled releases.

23. Implementation Checklist

- ☐ Repository and branching strategy established
- ☐ Database schema reviewed
- ☐ RBAC matrix approved
- ☐ Numbering sequences defined
- ☐ Company/branch isolation implemented
- ☐ Audit framework enabled
- ☐ File storage configured
- ☐ Notification adapters abstracted
- ☐ CRM workflows signed off
- ☐ Sales document lifecycle signed off
- ☐ Accounting posting matrix signed off
- ☐ HR policies configured
- ☐ Payroll formulas approved
- ☐ GST/tax configuration reviewed by qualified finance stakeholders
- ☐ Report reconciliation completed
- ☐ Backup and restore tested
- ☐ Security review completed
- ☐ UAT completed
- ☐ Production runbook prepared
- ☐ Admin/user documentation prepared

24. Final Engineering Notes

Treat accounting, payroll and authorization as high-integrity domains. Avoid placing financial logic only in UI code or database triggers. Centralize posting/calculation rules in testable backend services. Keep source documents and accounting entries linked through immutable references. Use configuration for company-specific policies, but do not make core accounting invariants configurable.

This specification is a baseline. Before development, the team should produce approved screen wireframes, an ER diagram, a permission matrix, accounting posting matrix, payroll policy sheet, API contract and sprint-level backlog.